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**BIT 2324 GEOGRAPHICAL INFORMATION
SYSTEMS**

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 9

Implementation Issues & Future of GIS

Learning outcomes

By the end of this topic you should be able to;

- *Define multilevel security*
- *Explain trusted systems*



9.1. Current options

Perhaps the first question asked by anyone when discovering GIS is what are the current options available ?. This question is often asked as directly as what is the best GIS ?. Quite simply, there is no best GIS. A wide variety of GIS software offerings exist in the commercial market place. Commercial surveys often are a good starting point in the assessment of GIS software. The number of GIS software offerings is approximately 10 if one eliminates the following:

- the university based research software, which tends to lack full integration and usually has narrow channels of functionality;
- the CAD vendors, who like to use GIS jargon but often



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cannot provide full featured functionality; and the consulting firms, that will provide or customize selected modules for a GIS but lack a complete product.

- One of the problems in evaluating the functionality of GIS software is the bias one gets from using one system or another. Comparing similar functions between systems is often confusing.
- Like any software, ultimately some do particular tasks better than others, and also some lack functionality compared to others.
- Any GIS should be evaluated strictly in terms of the potential user's needs and requirements in consideration of their work procedures, production requirements, and organizational context !



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- The experienced GIS consultant can play a large and valuable role in the assessment process.
- A current accepted approach to selecting the appropriate GIS involves establishing a
- benchmark utilizing real data that best represents the normal workflow and processes employed in your organization.
- Development of the benchmark should include a consideration of other roles within your organization that may require integration with the GIS technology. A logical and systematic approach as such is consistent with existing information systems (IS) planning methodologies and will ultimately provide a mechanism for a successful evaluation process.



9.2. Justification & expectations

GIS is a long term investment that matures over time. The turnaround for results may be longer term than initially expected. Quite simply, GIS has a steep learning curve. The realization of positive results and benefits will be not achieved overnight.

Even though the proper assessment of an appropriate GIS product requires a good

understanding of user's needs, most often systems are acquired based on less than complete

and biased evaluations. Nonetheless, even with the GIS in hand a properly structured and systematic implementation plan is required for a successful operation. Generally, a GIS implementation plan must address the following technical, financial,



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and institutional considerations:

- system acquisition tactics and costs;
- data requirements and costs;
- database design;
- initial data loading requirements and costs;
- system installation tactics, timetable, and costs;
- system life cycle and replacement costs;
- day-to-day operating procedures and costs;
- staffing requirements and costs;
- user training and costs; and
- application development and costs.



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Potential GIS buyers should be aware of the necessary investment required in hardware, software, training, supplies, and staffing. The cost of establishing a successful GIS operation is substantial. However, with realistic expectations and support the development of GIS within an organization that manipulates geographic data will almost certainly prove beneficial.

Certain considerations of data longevity, data capture, personnel hiring, etc. are the practical concerns of GIS implementation. The longer term implications, such as hardware/software maintenance and replacement, should also be considered. The acquisition of GIS technology should not be done without seriously considering the way in which GIS will interact with the rest of the organization.



9.3. Implementation Issues

The mere presence of an implementation plan does not guarantee success. Most organizations do not have sufficient staff to cope with the commitment and extra work required when introducing a GIS to existing operations. GIS implementation must also consider all technology transfer processes.

9.3.1. Common Pitfalls

Several pitfalls exist that most often contribute to the failure of a GIS implementation strategy. These are identified below:

- Failure to identify and involve all users Users in an operational GIS environment consist of operations, management, and policy levels of the organization. All three levels should be considered when identifying the needs of your



users.

- Failure to match GIS capability and needs. A wide spectrum of GIS hardware and software choices currently exist. The buyer is presented with a significant challenge making the right choice. Remember, the right choice will be the GIS that provides the needed performance no more, no less for the minimum investment. The success of a GIS implementation is particularly sensitive to the right hardware and software choices
- Failure to identify total costs. The GIS acquisition cost is relatively easy to identify. However, it will represent a very small fraction of the total cost of implementing a GIS. Ongoing costs are substantial and include hardware and software maintenance, staffing, system administration,



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initial data loading, data updating, custom programming, and consulting fees.

- Failure to conduct a pilot study The GIS implementation plan concerns itself with the many technical and administrative issues and their related cost impacts. Three of the most crucial issues, are database design, data loading and maintenance, and day-today operations. The pilot study will allow you to gather detailed observations, provided it is properly designed, to allow you to effectively estimate the operational requirements.
- Giving the GIS implementation responsibility to the EDP Department. Because of the distinct differences of the GIS from conventional EDP systems, the GIS implementation team is best staffed by non-data processing types. The



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specialized skills of the 'GIS analyst' are required at this stage. Reliance on conventional EDP personnel who lack these skills will ensure failure.

- Failure to consider technology transfer. Training and support for on-going learning, for in-house staff as well as new personnel, is essential for a successful implementation. Staff at the three levels should be educated with respect to the role of the GIS in the organization. Education and knowledge of the GIS can only be obtained through on-going learning exercises. Nothing can replace the investment of hands on time with a GIS

A critical requirement for all GIS implementations is that adequate education and training is provided for operational staff, as well as realistic priorities are defined with which to learn and



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apply the technology. This is where a formal training curriculum is required to ensure that time is dedicated to learning the technology properly. Adding GIS activities to a staff member's responsibilities without establishing well defined milestones and providing adequate time and training mechanisms is prone to failure. A focused and properly trained operations staff that has consistent training will result in greatly reduced turnaround times for operations, and ensure consistency in quality of product. The threshold point of the learning curve is typically around the two year time frame.

However, this is dependent on the ability of the organization to establish a well defined and a GIS acquisition based on well defined user needs and priorities is more likely to succeed than without. A major pitfall of most installations with GIS technol-



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ogy, e.g. particularly forestry companies and government agencies, is the lack of well defined user needs on which to base the GIS acquisition and implementation.

9.4. The Implementation Plan

GIS implementation requires a plan, and in each plan the following issue must be dealt with.

1. The implementation planning process: - organizing the tasks associated with implementation and smoothly making the transition from strategic planning to GIS implementation;
2. Creating an implementation management framework – Assigning roles and responsibilities to project participants.



3. Developing a conceptual overview: - Articulating the concepts envisioned at the awareness stage, a set of discrete parts that can be analysed, specified and implemented;
4. Managing expectations, establishing procedures and establishing an appropriate sequence of events; communicating the requirements for implementation to project participants in enough detail for all to understand what is to be done, how long it will take, what it will cost, and who will need to be involved and for how long.
5. Preparing the implementation plan: documenting what has been determined and agreed upon.
6. Developing “error free” specifications structuring the planning and design process to produce the fewest number of analysis and design errors.



7. Implementing the plan: successfully starting up and carrying through to completion.

9.5. Implementation Process

Implementation can be seen as a six phase process. They are:

1. Creating an awareness

GIS needs to be sold within an organization. The education of staff is very important. Depending on the way in which GIS technology is being introduced to the organization the process for creating an awareness may differ. Technical workshops are often appropriate when a top-down approach exists, while management workshops are often more relevant when a bottoms-up approach exists. Education of the new technology should focus on identify-



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ing existing problems within an organization. These often help justify a GIS acquisition. They include :

- spatial information is poorly maintained or out of date;
 - spatial data is not recorded or stored in a standard way;
 - spatial data may not be defined in a consistent manner, e.g. different classifications for timber information;
 - data is not shared between departments within an organization;
 - data retrieval and manipulation capabilities are inadequate to meet existing needs; and
 - new demands are made on the organization that



cannot be met with existing information systems.

2. Identifying System Requirements: The definition of system requirements is usually done in a user needs analysis. A user needs analysis identifies users of a system and all information products required by those users. Often a prioritization of the information products and the data requirements of those products is also undertaken. A proper user needs analysis is crucial to the successful evaluation of GIS software alternatives.

After user needs have been identified and prioritized they must be translated into functional requirements. Ideally, the functional requirements definition will result in a set of processing functions, system capabilities, and hardware requirements, e.g. data storage, performance. Experienced



GIS consultants often play a major role in this phase.

3. System Evaluations: Evaluating alternative hardware and software solutions is normally conducted in several stages. Initially a number of candidate systems are identified. Information to support this process is acquired through demonstrations, vendor literature, etc. A short listing of candidates normally occurs based on a low level assessment. This followed by a high level assessment based on the functional requirements identified in the previous phase. This often results in a rating matrix or template. The assessment should take into account production priorities and their appropriate functional translation. After systems have been evaluated based on functional requirements a short list is prepared for those vendors deemed suitable.



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A user-friendly system would include the following:

- System commands easily understood by the non-computer specialist;
- Efficient interaction for expert user. (command lines for faster interaction)
- Error messages should state clearly the nature of the error and probable its cause and direction to rectify the problem.
- A context sensitive 'help' facility at this stage

The second phase; standard benchmark, is then used to determine the system of choice. This is the administration of a standardized test procedure which provides a systematic means of comparing the performance level of competing systems. The buyer assesses how a system will



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perform under expected operating conditions. This operation starts with data that the vendor has not used and will actually be used in the GIS. A good system should perform those tasks specified as indicated without a specialist to change it along. A pilot study with an organizations data should be carried out to test the robustness of s system. This pilot study provides the necessary training for the staff and experience on the challenge of implementing a system. Potential benefits may also be realized at this stage. The pilot strictly should be carried out by the staff of the organization together with the consultations.

4. **Justifying the System Acquisition:** The proper justification of the chosen system requires consideration of several factors. Typically a cost-benefit analysis is under-



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taken to analyze the expected costs and benefits of acquiring a system. To proceed further with acquisition the GIS should provide considerable benefits over expected costs. It is important that the identification of intangible benefits also be considered.

The justification process should also include an evaluation of other requirements. These include data base development requirements, e.g. existing data versus new data needs and associated costs; technological needs, e.g. maintenance, training, and organizational requirements, e.g. new staff, reclassification of existing job descriptions for those staff who will use the GIS.

Some of the benefits of a GIS are:

- Better storage and updating of data



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- More efficient information retrieval
- More efficient production of information products
- Rapid analysis of alternatives
- The value of better decisions

5. **System Acquisition and Start Up:** After the system, e.g. hardware, software, and data, is acquired the start up phase begins. This phase should include pilot projects. Pilot projects are a valuable means of assessing progress and identifying problems early, before significant resources have been wasted. Also, because of the costs associated with implementing a GIS it is often appropriate to generate some results quickly to appease management. First impressions are often long remembered.

6. **Operational Phase:** The operational phase of a GIS im-



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plementation involves the on-going maintenance, application, and development of the GIS. The issue of responsibility for the system and liability is critical. It is important that appropriate security and transaction control mechanisms be in place to support the system. A systematic approach to system management, e.g. hardware, software, and data, is essential.

Revision Questions

EXERCISE 1. 🖱️ List the services of trusted computing.

Example 🖋️. Explain 5 components of trusted platform module.

Solution: i) Cryptographic keys, and session information.



EXERCISE 2.  What is a trusted computer system?



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Solutions to Exercises

Exercise 1. i) Authentication ii) Boot certification iii) Encryption

Exercise 1



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Exercise 2. A trusted computer system is a system that employs sufficient hardware and software assurance measures to allow its use for simultaneous processing of a range of sensitive or classified information.

Exercise 2